

Order Uropygi

Whip-scorpions

These flat-bodied arachnids have a whiplike tail (telson) at the end of their abdomen and can, in defense, squirt formic and acetic acids from a pair of abdominal glands. Their chelicerae resemble spider fangs. There is 1 family, with 106 species.

Family Thelyphonidae

Vinegaroons

Occurrence 100 spp. in North America and N.E. South America; in leaf litter, under stones, rotten wood



Vinegaroons hunt after dark, hiding in underground burrows during the day. They do not have a sting but can squirt

acid over 23½ in (60 cm), and can give a painful pinch with their pedipalps. Excluding the tail, they are up to 3 in (7.5 cm) long.

THELYPHONUS SPP.

Dark brown, with a tough, flattened body, these arachnids are found in Southeast Asia.

strong pedipalps for digging burrows



Order Amblypygi

Whip-spiders

Also known as tailless whip-scorpions, members of the 5 families (about 160 species) in this order do not sting or bite. They are nocturnal and cave dwelling, with fanglike, segmented chelicerae, large, shiny pedipalps, and an extremely long first pair of legs.

Family Phrynidae

Phrynids

Occurrence 60 spp. in tropical and subtropical regions; under stones, bark, in leaf litter and caves, among rocks



Brown, with dark markings, phrynids have a body ¾–2¼ in (0.5–6 cm) long. They scuttle rapidly sideways if disturbed. Females carry young for months in a sac under the abdomen.



PHRYNUS SPP.

This phrynid, from Central America, has bent its long front legs backward—a typical resting pose.

Order Opiliones

Harvestmen

Harvestmen—or daddy longlegs—are often mistaken for spiders, but they have rounded bodies, without a slender waist. Their legs are long and slender, and end in microscopic claws. Their mouthparts are small but they produce noxious secretions as a defense. Harvestmen are unusual among arachnids in that fertilization is direct. There are at least 6,125 species, in 48 families.

Family Cosmetidae

Cosmetids

Occurrence 700 spp. mainly in tropical regions of North and South America; under stones, among debris



Like most harvestmen, cosmetids are generally dull in color, although tropical species can be green or yellow, and a few can change color over a matter of weeks to blend with their surroundings. They range from ⅜–¾ in (5–11 mm) in length. Males tend to be smaller than females but have longer appendages and larger chelicerae. The eyes are located close together on a small “turret” called an ocularium. Females use their ovipositors to lay eggs in cracks and crevices in the soil.



VONONES SAYI

The Say's harvestman is found in Panama. It uses its front legs to smear toxic secretions on attackers.

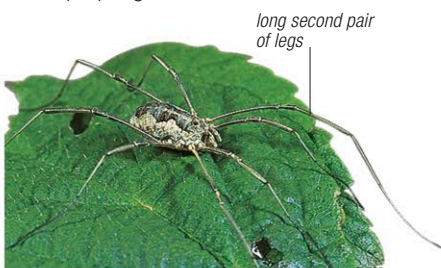
Family Phalangidae

Phalangiids

Occurrence 380 spp. worldwide; under stones, among leaf litter in woods, grassy areas



Phalangiids can have a legspan of up to 4 in (10 cm), but their soft bodies are rarely more than ⅜ in (1 cm) long. Like other harvestmen, their eyes are close together, and are on a “turret” above the rest of the body. Males and females differ; the enlarged chelicerae of the males are especially distinct. Females lay 20–100 eggs at a time, burying them in soil or under bark. Most species are nocturnal and prefer humid habitats. Phalangiids are both predatory and saprophagous.



PHALANGIUM OPILIO

Found in the Northern Hemisphere, the diurnal horned harvestman likes sunny spots, and inhabits woods, gardens, and grassland.

Order Acari

Ticks and mites

With over 48,200 species and about 438 families, ticks and mites are present in almost every habitat, particularly on land. They include many pests of crops and stored food, as well as parasites that attack mammals, birds, and reptiles. Most are less than ½ in (1 mm) long—although ticks can grow much bigger after feeding on blood—and unlike spiders, their bodies have no distinct divisions.

Family Ixodidae

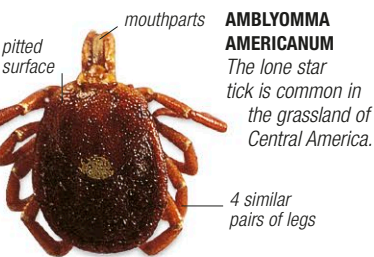
Hard ticks

Occurrence 700 spp. worldwide; in association with birds, mammals, and some reptiles



Hard ticks have a tough dorsal plate, which fully covers the males, but covers only the front half in females. The abdomen is soft and flexible to allow for large blood meals. The front part of their head projects forward and carries the mouthparts. Hard ticks are ⅛–¾ in

(0.2–1 cm) long but grow larger after feeding. Colors vary from yellow to red or black-brown; some species are highly marked. Some hard ticks carry viral diseases that affect humans, including encephalitis and Lyme disease, as well as diseases affecting cattle and poultry.



AMBLYOMMA AMERICANUM

The lone star tick is common in the grassland of Central America.

Family Trombididae

Velvet mites

Occurrence 300 spp. worldwide, especially in tropical regions; in or on soil, some associated with freshwater



Named after their dense, velvety “fur,” these bright red or orange mites eat insect eggs when they are adult, but

start life as parasites, attacking insects, spiders, and harvestmen. The adults often emerge from the soil after rain to mate and lay eggs.

TROMBIDIUM SPP.

Trombidium velvet mites are found in many parts of the world. This species, from South Africa, grows up to ⅜ in (1 cm) long.



Family Tetranychidae

Spider mites

Occurrence 700 spp. worldwide; on a range of trees, plants, and shrubs

Habitat All

These soft-bodied mites are orange, red, or yellow, and ⅛–½ in (0.2–0.8 mm) long. Many species

produce silk. Eggs are laid on leaves, and the young are protected by silk webs. They suck sap and heavy infestations can weaken a plant.



TETRANYCHUS SPP.

Abundant in temperate regions, red spider mites may produce 4–5 generations per year. They are pests of deciduous fruit trees.